Preliminary Assessment of a Multisite SpO_2 Probe for Clinical Application

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Abstract. Peripheral oxygen saturation (SpO_2) is a key non-invasive parameter for assessing systemic oxygenation, widely used in emergency, critical care, and outpatient settings. While fingertip pulse oximetry remains the clinical standard, its reliability is often compromised by motion artifacts, poor peripheral perfusion, skin pigmentation, hypothermia, and anatomical or environmental factors. These limitations are especially critical in patients with low SpO_2 levels, where conventional sensors may fail to detect rapid desaturation events. To address these challenges, this work explores alternative anatomical sites for SpO_2 measurement using a custom-designed probe. The device enables data acquisition from arterial territories such as the neck, earlobe, and wrist, offering a robust alternative when fingertip readings are unreliable. Preliminary findings demonstrate the feasibility of obtaining consistent SpO_2 signals from these locations under varied physiological and environmental conditions. This approach may improve monitoring accuracy in non-stationary scenarios and enhance the applicability of pulse oximetry in high-demand clinical environments.

Keywords: Oxygen Saturation, Multisite, Photoplethysmography.

1 Introduction

Peripheral oxygen saturation (SpO_2) is a critical clinical parameter for assessing cardiopulmonary function and systemic oxygenation. It is a non-invasive estimate of arterial oxygen saturation that serves as a vital sign in diverse medical settings, including emergency care, anesthesia, intensive care, and sleep medicine. For example, the rapid detection of hypoxemia ($SpO_2 < 90\%$) is essential to prevent organ dysfunction and life-threatening complications [1, 2]. The integration of photoplethysmography (PPG) into pulse oximetry has revolutionized continuous monitoring by leveraging optical absorption differences between oxygenated and deoxygenated hemoglobin, enabling real-time evaluation of oxygen-carrying capacity [3]. So, the clinical utility of SpO_2 lies in its ability to guide therapeutic interventions, triage decisions, and longitudinal management of conditions such as obstructive sleep apnea, chronic obstructive pulmonary disease, and acute respiratory distress, underscoring its indispensable role in modern healthcare.

Despite the widespread use of finger-based pulse oximetry for SpO_2 monitoring, its reliability is frequently compromised in real-world scenarios, limiting its clinical utility. Motion artifacts, which are common during patient movement, tremors, or physical activity tend to distort photoplethysmographic (PPG) signals leading to erroneous SpO_2 readings or complete signal dropout [1]. Additionally, poor peripheral perfusion, often observed in patients with different skin tone, hypothermia, hypotension, or vasoconstriction, reduces signal quality, as arterial pulsations become insufficiently detectable at the fingertip [2]. Challenges are further exacerbated in critically ill patients or those with low SpO_2 levels (<85%), where conventional devices may fail to capture rapid desaturations due to calibration biases or inadequate signal-to-noise ratios [3]. Finger-based sensors are also prone to displacement or improper contact during sleep or ambulatory monitoring, particularly in wearable applications, where long-term adherence is required [4]. Moreover, environmental factors such as ambient light interference or nail polish can introduce inaccuracies, while anatomical variations (e.g., thick skin, edema) alter optical path lengths, skewing SpO_2 calculations [5]. These limitations underscore the need for robust alternatives to traditional finger probes, particularly in non-stationary or resource-constrained settings. Recent advancements in PPG-based technologies aim to address these limitations, offering the possibility of portable solutions for ambulatory and ICU monitoring among others [4].

In view of the above, this work presents the exploration of alternative sites for SpO_2 evaluation. A probe-type device is proposed [6] in order to access different arterial territories, aiming to provide a practical solution to the needs and complexity imposed by clinical settings.

2 Materials and Methods

The nFISIO device consists of a pen-like probe module designed for the acquisition of PPG waveforms in arterial sites where superficial arteries are present. It integrates a PPG sensor equipped with multiple LEDs of different wavelengths, enabling accurate measurement of blood volume changes according to particular conditions. All hardware modules are housed within an ergonomically designed pencil-shaped enclosure, ensuring comfort and ease of use during the measurement. The device features wireless connectivity and rechargeable battery, allowing continuous and autonomous operation without action restrictions. It features a display screen (like a traditional oximeter) but supports sending signals and data to a proprietary application.

2.1 Main Core

The probe is managed by a microcontroller for real-time data acquisition, processing and transmission (STM32F103, STMicroelectronics, Amsterdam, The Netherlands). It has the capability of SPI, I2C connection, as well as analog inputs and digital outputs to ports, both to query variables such as battery voltage, the status of a push button for interaction with the user and the configuration of peripheral devices, such as the UART for sending information wirelessly. The system incorporates a 0.96-inch monochrome OLED display module (128×64 px, HOSYOND, China), connected via I²C interface for real-time visual feedback. This screen is used to provide essential operational information to the user during device activity. Due to its low power consumption, high contrast, and compact size, the OLED display is well suited for portable and wearable applications, offering continuous visibility of critical data without compromising energy efficiency.

2.2 Photoplethysmographic Activity Measurement and SPO2 Assessment

The MAX30102 module (Analog Devices, Massachusetts USA), whose sensor chip contains all its integrated power supply logic, was chosen to acquire the PPG signal. It is composed of red (650-670 nm) and infrared (870-900 nm) LED emitters. The maximum reflected intensity (minimum absorption/scattering) is obtained during the systolic phase, and the minimum intensity (maximum absorption/scattering) is obtained during the diastolic phase. This makes it possible to take measurements without exerting any pressure on the vessel and it can be easily coupled to peripheral and digital arteries [5]. SpO_2 is measured as follows: the sensor estimates the value by leveraging dual-wavelength reflected by 2 light-emitting diodes (LEDs), red (660 nm) and infrared (IR) (940 nm), to capture photoplethysmography (PPG) signals. These reflected captured waves contain information related to the distinct absorption properties of oxygenated (HbO_2) and deoxygenated hemoglobin (Hb), enabling the calculation of their relative concentrations. The evaluation is meant to study variations in signal amplitude between the LEDs. Consequently, normalization of their AC (pulsatile) and DC (baseline) components is required to enable meaningful comparisons. This process yields a parameter named as the "ratio of ratios" (R), defined as [5]:

$$R = \frac{AC_{Red}/DC_{Red}}{AC_{IRed}/DC_{IRed}} \quad ROS: Ratio\ of\ Ratios \qquad (1)$$

where AC_{Red} is the Absorbance Alternating Component red LED, DC_{Red} the Absorbance Continuous Component red LED, AC_{IRed} the Absorbance Alternating Component infrared LED and DC_{IRed} the Absorbance Continuous Component infrared LED. SpO_2 values can then be assessed by means of the linear approximation $SpO_2 = a - b * R$, which is derived empirically from R using calibration curves or lookup tables. These calibration datasets are typically generated through controlled studies involving diverse populations with some factors such as age, skin pigmentation, health status, and medical conditions that can influence measurement accuracy [2]. Usually, for practical applications, a linear approximation is chosen when R is in the range of 0.4–3.4. The selected approximation provides a simplified model, though curvilinear relationships are often employed for higher precision. Widely adopted in fitness and medical devices, this method balances non-invasive convenience with clinically acceptable accuracy under stable conditions. The coefficients “a” and “b” are chosen by designers according to their necessities. Then, the following formula is used:

$$SpO_2 = 110 - 25 * R \qquad (2)$$

Finally, PPG measurements are obtained from the main core from MAX30102 through the I2C, including sensor configuration parameters. After a stage of digital filtering (in order to obtain DC_{Red}, DCI_{Red} from AC_{Red}, ACI_{Red}), SpO_2 values are algorithmically obtained in terms of the described process.

2.3 Wireless Communication, Battery Control and Status Indicators

An HC-05 Bluetooth module (HC-Group) with "AT" command capability was used to allow parameter configuration directly from the device. To monitor the battery's maximum charge voltage of 7.4 V, an analog adaptation stage was implemented and connected to the microcontroller's ADC. This stage includes a resistive voltage divider and a transistor-based control mechanism, enabling periodic sensing rather than continuous monitoring, thereby improving energy efficiency. For visual feedback, two surface-mount RGB LEDs (Kingbright, Taiwan) were employed due to their low power consumption and reliable performance. One LED provides general device status indications, such as power state and battery level, using three color-coded alert levels, while the second LED indicates measurement activity and configuration modes. Additionally, a front-facing OLED display was integrated into the device to provide real-time visualization of physiological parameters, specifically peripheral oxygen saturation (SpO_2) and heart rate (HR). The OLED also presents a quantitative battery status indicator, offering more precise information on the battery charge level beyond the discrete LED alerts, thus enhancing usability and monitoring accuracy.

2.4 Housing

Custom ergonomic housing was developed to ensure both user comfort and optimal integration of internal components using additive manufacturing. The enclosure was designed to accommodate the main electronics, a rechargeable battery, and the PPG sensor module located at the distal end of the device. The design was carried out using a computer-aided design platform (SolidWorks, Dassault Systèmes SE, France), and the final components were manufactured using two distinct 3D printing technologies. The main structure was printed in ABS using a custom-built FDM 3D printer based on the Voron design (Siboor, China), selected for its ability to produce parts with a smooth surface finish and enhanced mechanical resistance. The section housing the PPG sensor was printed using a resin printer (Photon, Anycubic, China), allowing for high-resolution features and superior surface smoothness, which are particularly important for ensuring good optical coupling and comfortable skin contact. In addition, the enclosure incorporates localized wall thinning to allow internal indicator LEDs, such as those signaling battery status, to be visible externally without requiring openings, thus preserving structural integrity.

2.5 Data Management

The nFISIO probe enables SpO_2 measurement and can output data in either processed (displayed on screen) or raw formats (for alternative processing). To support this, a complementary application was developed using the Qt multi-platform framework® (v5.13) along with the QCustomPlot library for real-time visualization. Both raw and processed data are stored in .csv format files.

2.6 SpO_2 Evaluation Protocol

Sixteen (16) individuals (23±3 years), with no history of vascular disease, participated in the probe evaluation. SpO_2 measurements were performed in the following arterial territories: neck (carotid artery), earlobe, upper arm (brachial artery), wrist (radial artery) and fingertip, with the individuals in the supine position. Research protocol was carried out in accordance with the Code of Ethics of the World Medical Association (Declaration of Helsinki).

2.7 Statistical Analysis

Fingertip measurements were first compared with those obtained using a commercial finger-probe pulse oximeter (Contec Medical Systems Co., Ltd). Mean absolute error (MAE), mean square error (MSE), and root mean square error (RMSE) were then calculated. Finally, a one-way ANOVA test with Bonferroni correction was applied to determine whether significant differences existed among the measurements obtained from each arterial territory.

3 Results

Figure 1 shows the nFISIO probe, whose design enables the evaluation of PPG wave signals at multiple sites in the arterial vasculature. The upper section of the screen is structured as a status bar, showing a quantitative battery level indicator (expressed as a percentage) along with the current operating mode. Below this area, the display is divided into two main fields, dedicated to showing real-time physiological measurements: SpO_2 and HR. Both values are shown as mean ± standard deviation, according to the evaluated heart beats.

Fig. 1. The nFISIO probe displaying real-time SpO_2 and HR measurements.

The implementation of the evaluation protocol is shown in Figure 2. It can be seen as an example, the measurement process on one subject. The complementary software application is also shown for data storage and post-processing purposes.

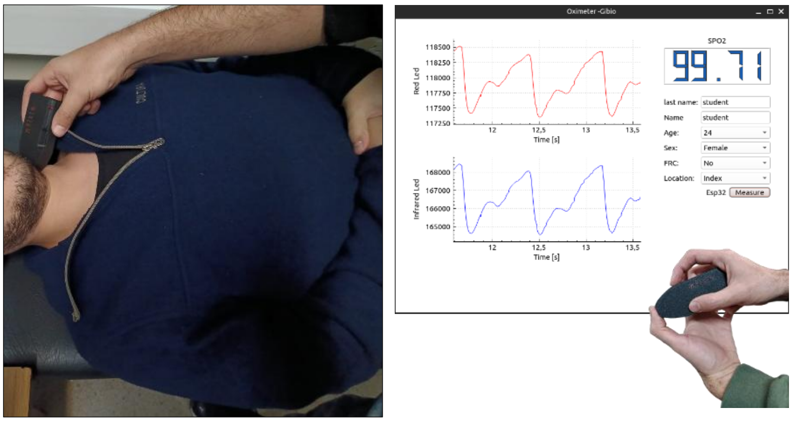

Fig. 2. Left: Experimental setup showing nFISIO data acquisition in different arterial places. Right: Complimentary graphical user interface of the SpO_2 acquisition system. The left panels display real-time PPG signals from red and infrared LEDs, while the right panel includes user input fields (e.g., age, sex, measurement site), device selection, and a digital readout of the estimated SpO_2 value.

MAE, MSE, and RMSE values obtained from the comparison with a commercial fingertip device were 1.51%, 2.98%^{2}, and 1.73%, respectively. Subjects' SpO_2 measurements are summarized in Table 1, categorized by arterial site. Values are expressed as mean ± standard deviation. Both the finger and earlobe sites showed the highest SpO_2 readings.

Table 1. Subjects' SpO_2 measurements.

Age [years]	Arterial Territory	SPO2 Value [%]
	Fingertip	99.35 ± 0.68
	Wrist	95.32 ± 3.20
23 ± 3	Upper arm	94.54 ± 3.06
	Neck	95.44 ± 4.13
	Earlobe	97.31 ± 2.27

Figure 3 shows boxplots corresponding to the measurements performed at each arterial territory. The results highlight variability in SpO_2 values depending on the measurement site, with the highest median observed at the fingertip and the lowest at the forearm. Statistically significant differences were particularly observed between fingertip measurements and those from all other arterial territories.

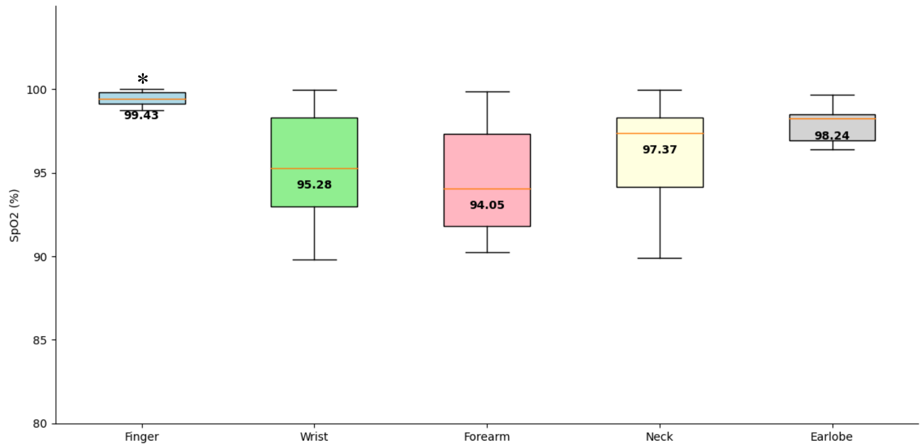

Fig. 3. Boxplot distribution of SpO_2 values by measurement site. The fingertip shows the highest median SpO_2 (99.34%), while the neck presents the lowest (92.29%). Significant differences were found between fingertip and the rest of arterial territories (*$p<0.05$).

4 Discussion

In this work, alternative anatomical sites were explored for acquiring SpO_2 measurements. To this end, a probe-type device was developed for use in clinical settings. Motion artifacts, such as patient movement, tremors, physical activity, as well as poor peripheral perfusion due to factors like skin tone, hypothermia, hypotension, and environmental conditions, often distort PPG signals. This can lead to inaccurate SpO_2 readings or complete signal dropout when measuring at standard sites [1]. The results verified the device's capability to obtain reliable SpO_2 measurements from alternative arterial locations such as the neck, earlobe, and wrist, suggesting a promising option when conventional finger-based measurements fail.

Based on the reviewed literature, hospital settings face significant limitations when conventional finger-based pulse oximetry is unreliable or unobtainable due to poor peripheral perfusion (e.g., hypothermia, hypotension, vasoconstriction), motion artifacts (e.g., tremors, agitation), anatomical constraints (e.g., burns, edema, nail pathologies), or physiological factors like darker skin tones or nail polish. These conditions cause signal dropout, erroneous SpO_2 readings, or complete measurement failure in up to 80% of critical cases, necessitating invasive arterial blood gas (ABG) analysis for verification. Clinicians often resort to off-label probe placement (e.g., ear, nose) using finger sensors, but studies validate severe inaccuracies. [7] demonstrated that 80% of ear SpO_2 values overestimated SpO_2 by ≥ 3% due to optical shunting and tissue heterogeneity. While solutions like forehead sensors, multi-wavelength systems, and accelerometer-based motion cancellation [8] have been proposed, they remain vulnerable to site-specific artifacts (e.g., venous pulsation in Trendelenburg positioning or swallowing-induced noise in neck PPG [9]). Consequently, a critical unmet need persists for a versatile probe enabling accurate, motion-tolerant SpO_2 measurements across diverse anatomical sites and clinical scenarios.

The proposed device demonstrated good results and a promising performance comparable to validated alternative-site pulse oximetry systems while addressing critical clinical limitations. When considering other pulse oximetry systems, it is possible to distinguish those such as: a wrist-based reflectance oximetry system by [2] that achieves ISO/FDA-compliant accuracy (A_{RMS} = 2.7-3.4%) when incorporating robust motion/contact quality indices [2] surpassing ear measurements which show 80% error rates ≥3% due to optical shunting [7]. Notably, its performance during sleep (279 hours of valid data across 54 subjects) parallels forehead sensor reliability [10] while resolving fingertip obtrusion issues. Though neck PPG exhibits strong SpO_2 correlation of r=0.82 [9]. Green-light wrist PPG [4] fails during ambulation, while conventional finger sensors discard 17-23% of data from perfusion-compromised patients [3].

Our device appears to mitigate swallowing artifacts through multi-site redundancy. It is noteworthy that no existing system combines this cross-anatomical adaptability (finger/wrist/neck/forearm/ear) with motion resilience. By enabling site-specific optimization (e.g., avoiding venous pulsation-prone areas during Trendelenburg positioning), the probe overcomes singular-location constraints observed in prior studies. The pencil-type design represents a structural innovation unreported in existing literature, contrasting with rigid wrist-worn devices [2], flat forehead patches [10], or coin-shaped neck sensors [9]. Its cylindrical form factor uniquely enables cross-anatomical adaptability (finger/wrist/neck/forearm/ear) while maintaining motion-resilient skin contact, solving site-specific limitations like venous pulsation during Trendelenburg positioning or swallowing artifacts that compromise single-location systems.

When analyzing the obtained SpO_2 measurements, a systematic decrease is observed with respect to fingertip reference values which usually are 97% at finger and 90–94% at neck/wrist. This seems to be due to site-specific vascular anatomy and tissue optics. A substantial anatomical difference was observed between men and women, involving deeper arterial pathways at alternative sites, as suggested by the data. It is proved that there exists 40% greater radial artery depth in female wrists. As a consequence, this anatomical difference might tend to increase photon scattering and to reduce AC/DC signal ratios by 18–82% [2, 9]. With respect to other measurement influences, it is noticed that venous pulsations depend on different positions where measures are taken (e.g., Trendelenburg). Thereby, further contamination can be observed on PPG waveforms since hydrostatic pressure changes occur. Additionally, optical shunting on curved surfaces (ear/neck) could be distorting ROS calculations. These biophysical challenges must be addressed by site-specific calibrations due to "normal" SPO2 intrinsically varies depending on arterial accessibility, tissue depth, and gender physiology.

While reflectance PPG technology offers promising alternatives to conventional finger pulse oximetry, its accuracy remains constrained by site-specific anatomical and optical factors when compared to the finger-index reference (consistently 97–99%). Notably, although the ear and wrist provide improved patient comfort, their measurements exhibit systematic deviations: ear sites frequently overestimate SpO_2 by ≥3% due to tissue heterogeneity and optical shunting [7], whereas wrist readings, particularly in females with deeper radial arteries, show signal attenuation requiring 17–23% data rejection to achieve ISO-compliant accuracy [2]. Furthermore, neck PPG demonstrates strong correlation of r=0.82 [9] but suffers from erratic reliability due to motion artifacts (e.g., swallowing), hair/beard interference, and venous pulsation during positional changes. Critically, all alternative sites reveal inherent biophysical limitations such as arterial depth variations, perfusion disparities, and photon scattering collectively causing 3–7% absolute deviations from fingertip values. Consequently, larger validation cohorts (>200 subjects per FDA guidelines) are imperative to establish robust site-specific calibration curves, especially given by current studies, limited sample sizes (n≤54) and unresolved environmental confounders.

In light of the results obtained, the nFISIO probe could be adopted for oxygen saturation determination in situations where readings at traditional sites fail. Future validation tests are required to assess the feasibility of implementing the device in hospital settings.

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